

How Colonization Economies Work

A player-facing guide for **Elite Dangerous** CMDRs using Raven Colonial to plan colonization ports, Odyssey settlements, hubs, and installations. It explains the same model as the technical economy model (companion: `economy-model.md`), but without code terms.

What you see in Raven Colonial: every dockable port, settlement, hub, or installation with a market can show percentages for Agriculture, Refinery, Military, High Tech, Tourism, and the other economy types. A value like **225%** means the market has economy strength **2.25** in the data.

Those percentages come from three places:

- The site's **host location**: planet type, moon type, star, asteroid cluster, Biological/Geological signals, volcanism, rings, and similar conditions.
- The system's **reserve level**: Pristine, Major, Low, Depleted, and related resource states.
- **Market links** from other valid builds in the same system.

Spansh check: for completed in-game stations, Raven Colonial can compare its estimate with Spansh/EDSM market snapshots. Spansh is comparison-only; it does not change the calculation.

Surface and Orbital Terms

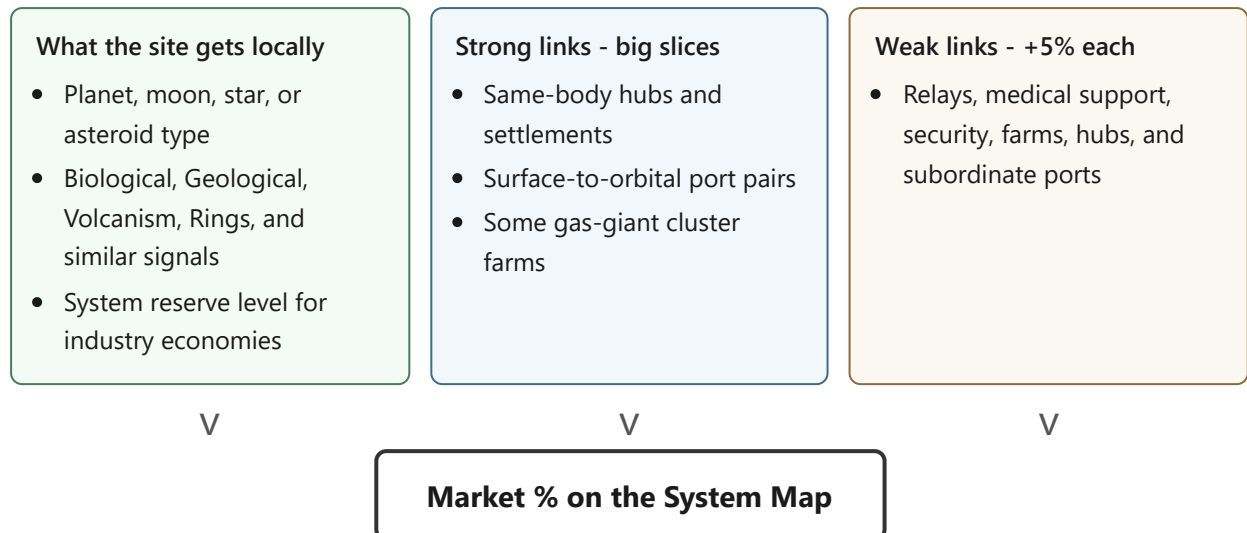
Raven Colonial uses these terms throughout this guide:

Term	Meaning
Surface port	A dockable starport or outpost built on the surface of a landable planet or moon
Orbital port	A dockable starport or outpost in space, including ports around a planet, moon, star, or asteroid cluster
Settlement	An Odyssey surface settlement
Hub / installation	A supporting build such as a relay, security post, space farm, comms installation, refinery hub, or science hub
System primary port	The first port the game has you build when you colonize a system. It should only be changed in Raven Colonial if that primary port is wrong
Body primary port	The main local market-link receiver for a body. When both orbital and surface ports share a body, the orbital port usually leads the pair
Subordinate port	An extra port attached under a body primary port or hub

When this guide says **same body**, it means builds on the same planet, moon, star, or asteroid cluster. When it says **other bodies**, it means different locations in the same system.

The Big Picture

Think of each market row as a ledger. Raven Colonial adds every line that applies, then shows the total.



Mechanism	Typical amount	Do local conditions change it?
Local row	+100%, +50%, or +40% chunks	Yes
Strong link	+40%, +80%, or +120% per source	Yes, especially Agriculture and reserve-level industry
Weak link	+5% per source	No, weak links stay flat

Market Links shows candidates that can link into a port. The audit table shows which candidates actually changed the market row.

Who counts: by default, only builds marked **Complete** affect the economy model. **Use all Sites** adds planned and in-progress builds for what-if planning, but invalid rows still stay below **BROKEN BELOW**.

What To Build For Each Economy

This is the short CMDR version. Later sections explain the edge cases.

Goal	Typical helpers
Agriculture	Biological bodies, Earth-like / Water worlds, space farms, Agriculture settlements
Refinery / Industrial / Extraction	Rocky, Rocky-Ice, Icy, HMC, Metal-rich, rings, Geological / Volcanism, refinery / industrial / extraction hubs
High Tech	Relays, medical installations, scientific settlements/hubs, Ammonia / Earth-like / Biological / Geological bodies
Military	Security installations for weak links, military hubs for local strong links, star-based ports
Tourism	Tourist builds, Earth-like / Water / Ammonia bodies, Biological / Geological signals, black holes, neutron stars, white dwarfs

What Counts In The Model

The System View has two calculation modes:

Mode	What counts
Completed sites only	Complete sites only
Use all Sites	Complete, building, and planned sites up to the cut line

Rows with missing or unsafe modeling data are forced below **BROKEN BELOW** and never feed the model:

- Unknown body
- Missing, `unknown`, or `null` build type
- Positive single-digit `marketId` values (1 through 9)
- Demolished site

A new or planned site with placeholder `marketId: 0` is not automatically broken. If it has a valid body and build type, it belongs below the normal cut line in completed-only mode.

Ports, Settlements, Hubs, And Installations

Dockable ports are the main market rows most CMDRs care about. General colony ports use the full model: local row, strong links, weak links, then primary economy selection.

Surface ports and **Orbital ports** can behave differently when they share a body. If a Surface colony port and an Orbital colony port are both present, the Surface port can push its local economies into the Orbital port as a strong link.

Specialized ports start with their specialty:

Specialized port location	Starting specialty
Surface	+50%
Orbital	+100%

Examples: Refinery outpost, Tourism starport, Extraction outpost.

Odyssey settlements have a shorter path. They get a fixed settlement economy, local buffs, and any settlement floors. Strong and weak links do **not** apply to the settlement's own market row, but settlements can still send links outward when the link graph makes them a source.

Hubs and installations mostly feed links into ports. If they have their own market economy, Raven Colonial gives them a fixed facility row instead of running the full port model. Link-only facilities do not get their own market percentage.

How A Colony Port Gets Its Numbers

1. Local body or orbital location

The port starts with economies from where it is built. Rocky bodies add Refinery, Metal-rich and HMC bodies add Extraction, Earth-like worlds add Agriculture / High Tech / Military / Tourism, stars add Military, black holes / neutron stars / white dwarfs add High Tech and Tourism, and asteroid clusters add Extraction.

2. Signals and reserve level

Biological and Geological signals add local rows where they apply. In **Pristine** or **Major** systems, Refinery, Industrial, and Extraction can gain **+40%**. In **Low** or **Depleted** systems, those same industry rows can lose **-40%** on non-settlement rows.

3. Strong links

Same-body facilities, hub children, settlements, and Surface-to-Orbital port pairs can add large economy slices into a primary port.

4. Weak links

Eligible sources add **+5%** each. Weak links are intentionally small and do not get extra Pristine, Biological, Terraformable, tidal, or icy modifiers.

5. Primary economy

The highest total becomes the site's **primary economy**, which is the main economy tag the game highlights. Other non-zero economies still remain on the market row.

Local Rows And Body Intrinsic

Common body/feature rows:

Body or feature	Common local economy
Earth-like world	Agriculture, High Tech, Military, Tourism
Water world	Agriculture, Tourism
Ammonia world	High Tech, Tourism
Gas giant / water giant	High Tech, Industrial
HMC / metal-rich	Extraction
Rocky-Ice	Industrial, Refinery
Rocky	Refinery
Icy	Industrial
Asteroid cluster	Extraction
Star	Military
Black hole / neutron star / white dwarf	High Tech, Tourism
Rings	Extraction
Biological signal	Agriculture and Terraforming
Geological signal	Extraction and Industrial

Reserve level affects Extraction, Industrial, and Refinery:

Reserve level	Effect
Major / Pristine	+40%
Low / Depleted	-40% on non-settlement own rows

Other common local buffs:

- Volcanism can add **+40% Extraction**.
- BIO/GEO or ELW/AW can add **+40% High Tech**.
- Black holes, neutron stars, white dwarfs, BIO/GEO, and ELW/WW/AW can add **+40% Tourism**.

Strong Links

Strong links are the main way to specialize a port. They are large because they represent local infrastructure directly feeding the port.

Strong Link Size

Strong link size follows **construction tier**, not pad size.

Source tier	Strong link contribution
T1	+40%
T2	+80%
T3	+120%

A T2 bio research settlement is a +80% strong-link source even if its pad is small.

Surface + Orbital Pairs

If a Surface colony port and an Orbital colony port share the same planet or moon, the Surface port's local economies can strong-link into the Orbital port.

Body primary behavior is calculated from the current valid site order for that body. The system primary port remains the original port the game had you build first for the system.

Hub Children

A hub can bring its children along as **sub-strong** links. For example, a Refinery hub with settlements underneath can show the hub plus each child as strong-link contributors into the primary port.

In a **Pristine** or **Major** system, each Refinery / Industrial / Extraction strong-link contribution can also receive its own +40% reserve boost.

Gas-Giant Cluster Farms

A space farm on a sibling moon under the same gas giant can strong-link Agriculture into that moon's **primary** port only. If a second port exists on that moon, it gets Agriculture through weak +5% steps instead of the farm's strong link.

Weak Links

Weak links are background support from elsewhere in the system. Each applied weak link is +5%.

Who Sends Weak Links

Sender	What it sends
Relay installation	High Tech +5% to eligible ports on other bodies
Medical installation	High Tech +5% support, and can help relays count for starports
Security installation	Military +5% to eligible ports on other bodies
Demeter space farm	Agriculture +5% when it is a qualifying space farm
Economy-bearing hub	Its primary economy outward
Subordinate starport / outpost / hub	Its primary economy outward
Colony body primary	Agriculture outward through the main weak-link path
Primary port at the central star	Agriculture outward only; Military and other local rows stay on that port

Relay And Medical High Tech

Relays are straightforward on outposts: if the relay is eligible and it is not on the same body as the receiver, it adds **+5% High Tech**.

Starports are pickier. A relay adds **+5% High Tech** to a starport only when the starport has a High Tech reason to receive it. That reason can be:

- High Tech already exists on the starport's market row, or
- Another non-relay High Tech weak source is linked, such as medical or scientific support.

If a starport has no High Tech row and no non-relay High Tech support, the relay may still appear in **Market Links**, but it does not add +5% to the economy total.

Agriculture From Distant Stars

A Surface colony with no local Agriculture may accept only **one** Agriculture weak link per distant star's subtree. Farms and Agriculture ports around another star count once for that receiver. Agriculture settlements are exempt from that distant-star cap.

Agriculture

Agriculture is the special case because local agriculture, strong-link agriculture, and weak-link agriculture use different rules.

Local Agriculture Row

On the site's own market row:

- Biological signals can add **+100% Agriculture** if the world type does not already grant Agriculture.
- Biological, Earth-like, and Water world conditions can add **+40%** Agriculture buffs.
- The optional **Terraformable Agri Bonuses** button can add a +40% what-if buff.
- Icy bodies or tidal conditions can apply **-40%** Agriculture. Settlements only receive that negative row when a positive Agriculture body buff was applied.

Agriculture Strong Links

Strong-link Agriculture uses the receiver's conditions:

Receiver condition	Strong-link Agriculture effect
Biological	+40%
Earth-like / Water world	+40%
Terraformable toggle enabled	+40%
Icy / Rocky-Ice	-40%
Tidal to the star	-40%

A tidal or icy penalty cannot reduce an Agriculture strong link below **+10%**.

Agriculture Weak Links

Weak Agriculture is always **+5%** per applied farm, Agriculture port, or Agriculture settlement source. These +5% steps do not receive Biological, Terraformable, icy, tidal, or reserve modifiers.

Most ports can receive any number of valid weak Agriculture links. Tidal orbital cluster colony ports are the notable capped case:

Tidal orbital cluster case	Weak Agriculture budget
Same-body Agriculture facility/settlement strong link exists	55%
No same-body Agriculture strong link	65%

Terraformable Agri Bonuses

The System page has a **Terraformable Agri Bonuses** button. It is off by default because current Elite Dangerous market behavior does not consistently confirm the expected Terraformable Agriculture modifier.

Turn it on to preview what Agriculture numbers would look like if that in-game behavior is fixed later. The setting is saved in your browser.

Refinery, Industrial, And Extraction In Pristine Systems

In **Pristine** or **Major** systems, reserve level matters in two places:

Where	Reserve effect
Port's local Refinery / Industrial / Extraction row	+40% once
Each Refinery / Industrial / Extraction strong link	+40% per strong-link contribution

In **Low** or **Depleted** systems, the same rows can receive **-40%** instead.

Example: Orbital starport over a rocky moon in a Pristine system.

Ledger line	Amount
Rocky body gives Refinery	+100%
Pristine buff on local row	+40%
Strong link from Surface colony partner	+40%
Pristine buff on that strong link	+40%
Sub-strong link from Refinery hub child	+80%
Pristine buff on that sub-strong link	+40%
Total Refinery	340%

High Tech and Tourism body bonuses on strong links can apply as their own contributions. Reserve-level industry boosts are especially visible because they stack per Refinery / Industrial / Extraction strong-link contribution.

Multiple Ports On One Body

Frontier's colonization rules make multiple dockable ports around one planet, moon, star, or asteroid cluster awkward. Raven Colonial models the behavior this way:

1. The model chooses a body primary from valid calculated sites.
2. Orbital ports normally outrank surface ports for the same body.
3. Other same-body ports become subordinates.
4. Subordinates can send their primary economy outward as weak support.
5. Same-body partner ports can create strong links into the body primary.

The Order for Calculations panel groups rows by body for readability, but the math does not depend on arbitrary visual row dragging.

Tier Points And The Order Panel

Tier-point math uses a canonical tax order so the grouped display cannot change the result.

1. Skip the system primary port.
2. Consider valid starports with tier requirements.
3. Sort higher tier before lower tier.
4. Sort by body order.
5. Sort orbital before surface.
6. Sort by market/name fallback.

The Order for Calculations panel is now informational:

- **Completed sites only** and **Use all Sites** mirror the main System View toggle.
- **Group by body** regroups the visible information without changing the model facts.
- **Auto re-order markets** applies current Spanish inversion hints only.
- Users no longer drag table rows.
- **Change Primary Port?** is a correction tool for cases where Raven Colonial has the wrong system primary port.

Warnings about insufficient Tier 2/Tier 3 points appear for planned sites. Complete and building sites are already locked in.

Broken Below

Rows below **BROKEN BELOW** are visible but excluded. This protects the model from imports or stale data that cannot be safely interpreted.

Broken reason	Why it is excluded
Unknown body	Body order, local rows, and body links cannot be trusted
Missing/unknown/null build type	The site has no valid economy pipeline
Positive single-digit market id	Treated as bad imported placeholder data
Demolished	The site should not affect current calculations

Valid planned sites belong below the normal cut line in completed-only mode, not below **BROKEN BELOW**.

Spansh Compare And Market Inversions

The compare panel tries to match completed dockable sites against Spansh:

1. Use journal/Raven Colonial `marketId` if it resolves to an operational Spansh economy row.
2. Fall back to EDSM station-name matching when the journal id is missing, stale, or a construction placeholder.

Sites with no landing pads are treated as lower-reliability comparisons because their public market ids can remain stuck on construction-era snapshots.

If two same-body sites appear swapped in Spansh, Raven Colonial can show up/down hints in Order for Calculations. **Auto re-order markets** applies those recommended market inversion swaps only; it does not perform a general site regroup.

System Effects Buff/Nerf

Raven Colonial applies the system effects model by default. A commander/developer setting can still disable it for testing comparisons.

When enabled, the system primary port receives buffs while other facilities receive nerfs for affected system stats:

Stat	Primary	Non-primary
Development	+40%	-10%
Security	+40%	-10%
Standard of Living	+40%	-20%
Technology	+20%	-25%
Wealth	+40%	-25%
Population / Max population	unchanged	unchanged

Quick Answers

Question	Answer
How big is a weak link?	Always +5%.
How big is a strong link?	+40%, +80%, or +120% by source tier.
Does Spansh change calculations?	No, it is comparison-only.
Can users drag the calculation order?	No. The panel is informational; primary port correction is explicit.
What is the system primary port?	The first port the game has you build when you colonize a system.
Do invalid imported sites count?	No. They go below BROKEN BELOW .
Is <code>marketId: 0</code> broken by itself?	No. Positive single-digit ids <code>1</code> through <code>9</code> are the bad-id case.
Are Agriculture weak links usually capped?	No. Tidal orbital cluster colony ports are the capped case.